

Digital System Design

LAB VI

Stepper Motor

DONG Yunyang

dongyy@sustech.edu.cn

411, No. 2, Hui Yuan

Tencent Meeting: 874-068-9694

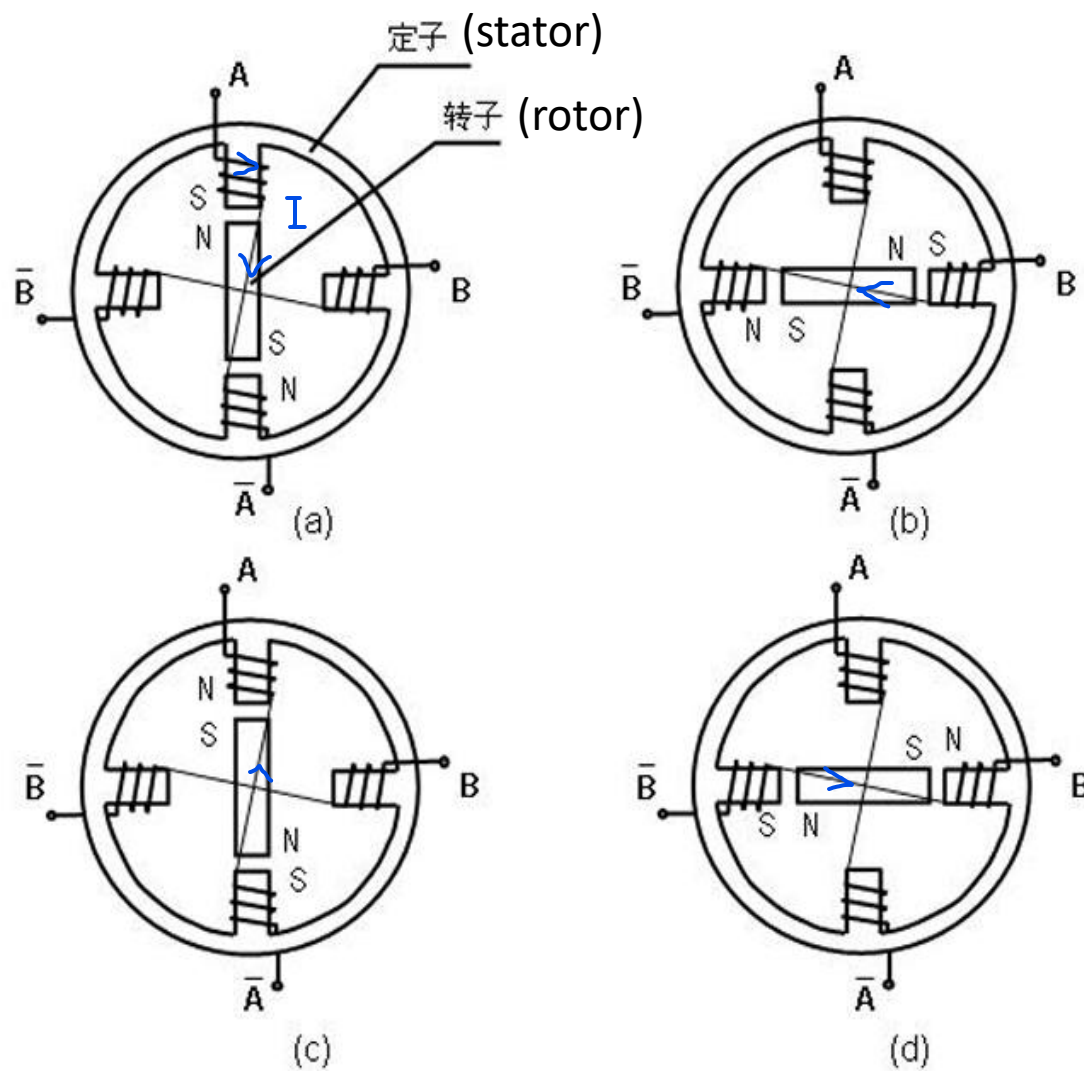
Stepper Motor

步进电机是一种将电能转换为机械能的装置。步进电机每接收到一个脉冲，就会向前转动一步。步进电机能够在没有反馈机制的情况下精确控制电机的位置，通常用于数字控制系统当中。

型号	28BYJ-48
28	外径28mm
B	步进电机
Y	永磁式
J	减速型（减速比1:64）
48	四相八拍
步进角度	5.625° /64 ($360/5.625*64 = 4096$ 个脉冲转一圈)
工作电压	5V DC



Simple Model



$A \rightarrow \bar{A}$

$B \rightarrow \bar{B}$

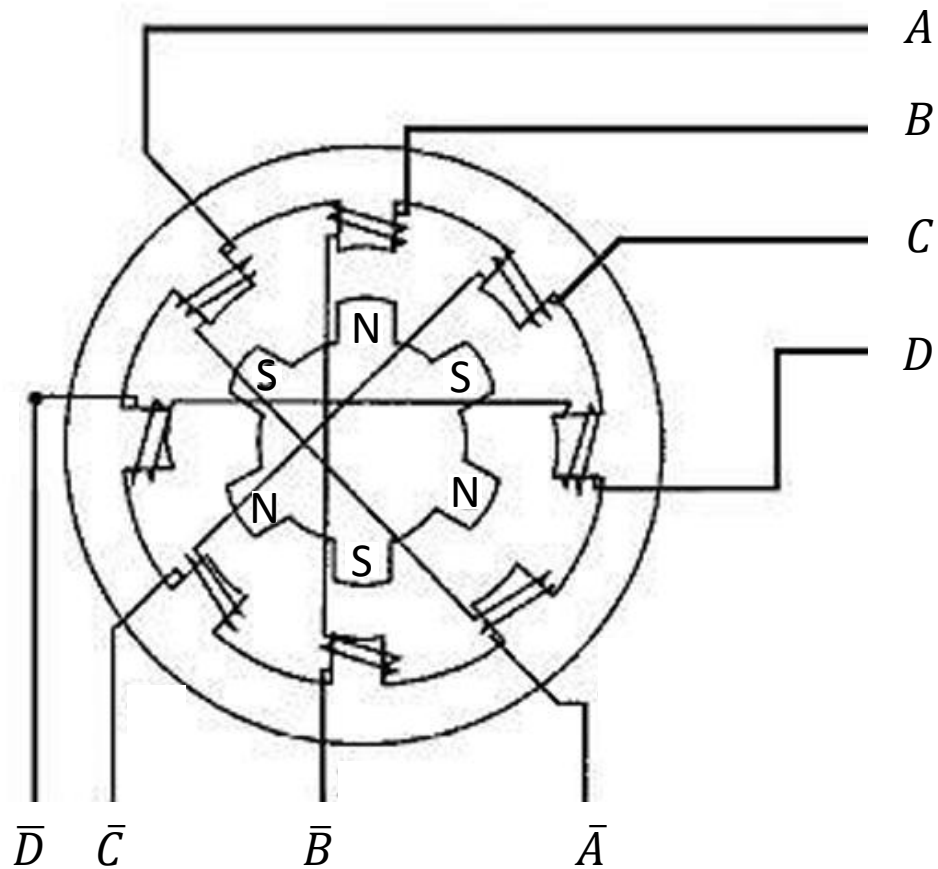
$\bar{A} \rightarrow A$

$\bar{B} \rightarrow B$

4个定子齿
步进 $360/4 = 90^\circ$

4线

Simple Model

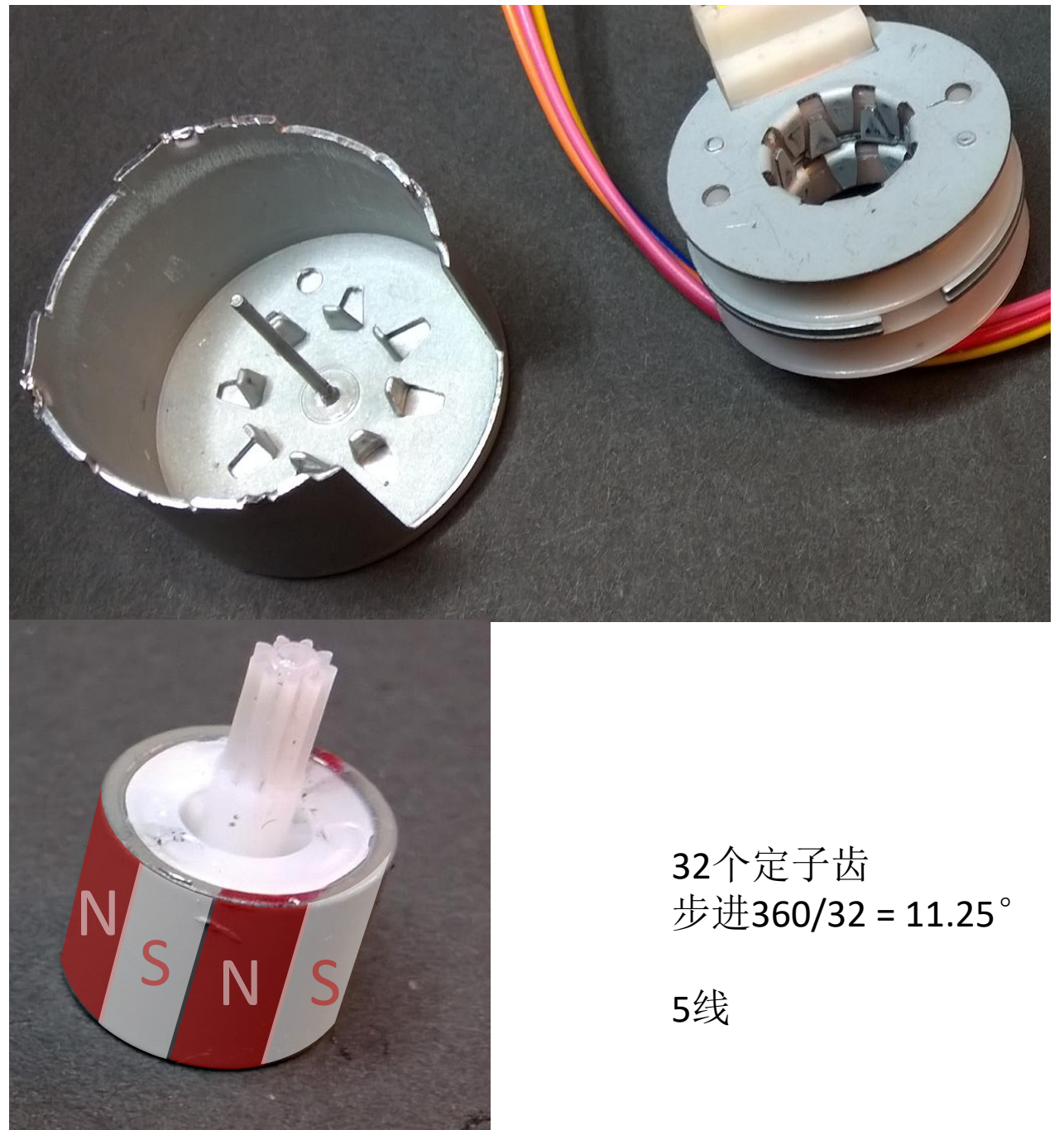
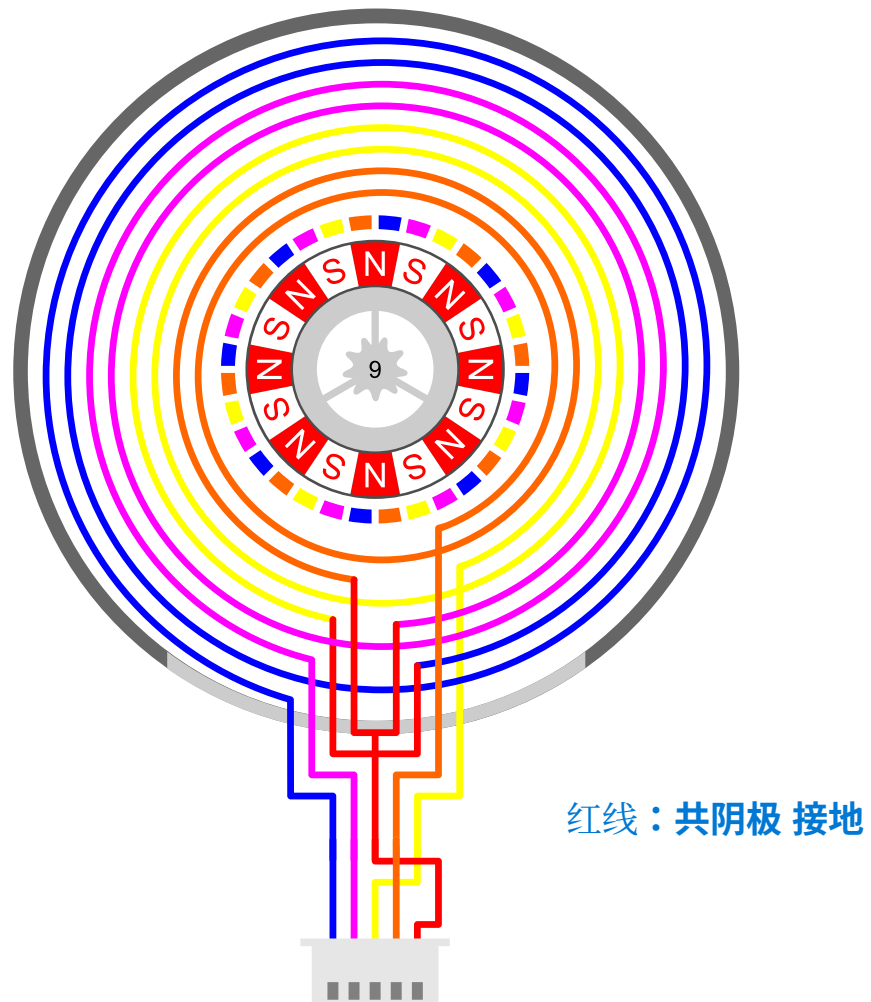


$A \rightarrow \bar{A}$
 $B \rightarrow \bar{B}$
 $C \rightarrow \bar{C}$
 $D \rightarrow \bar{D}$
 $\bar{A} \rightarrow A$
 $\bar{B} \rightarrow B$
 $\bar{C} \rightarrow C$
 $\bar{D} \rightarrow D$

8个定子齿
步进 $360/8 = 45^\circ$

8线

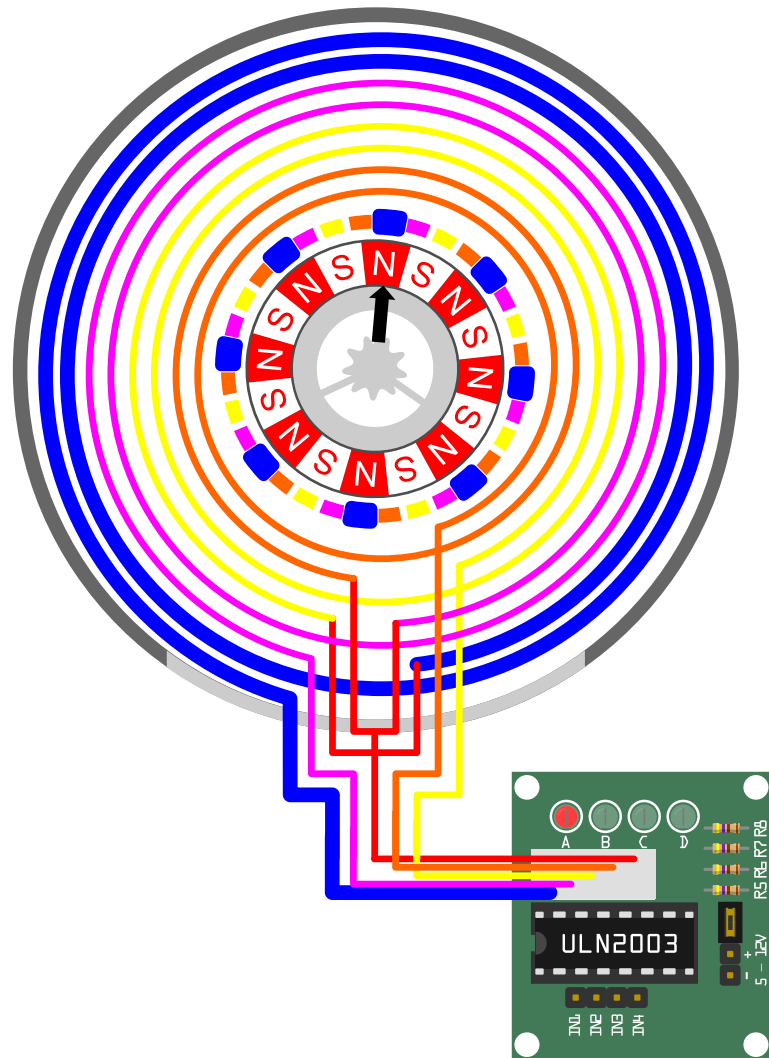
Structure



32个定子齿
步进 $360/32 = 11.25^\circ$

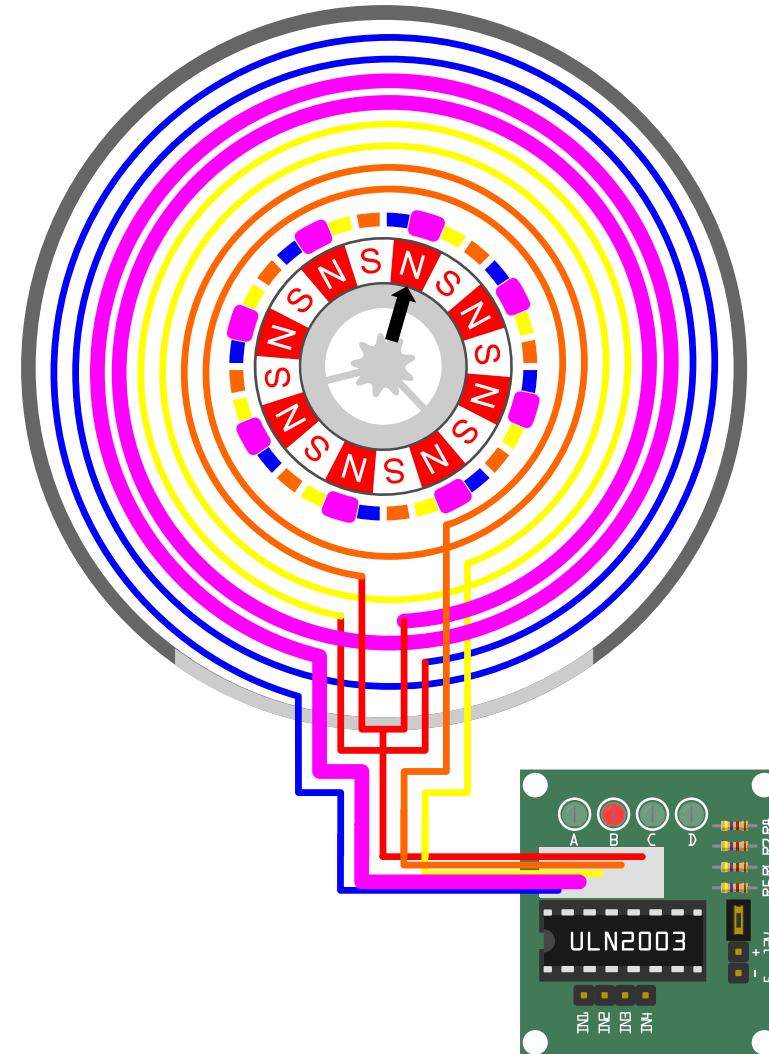
5线

Driving Mode – Wave Driving



蓝色高电平

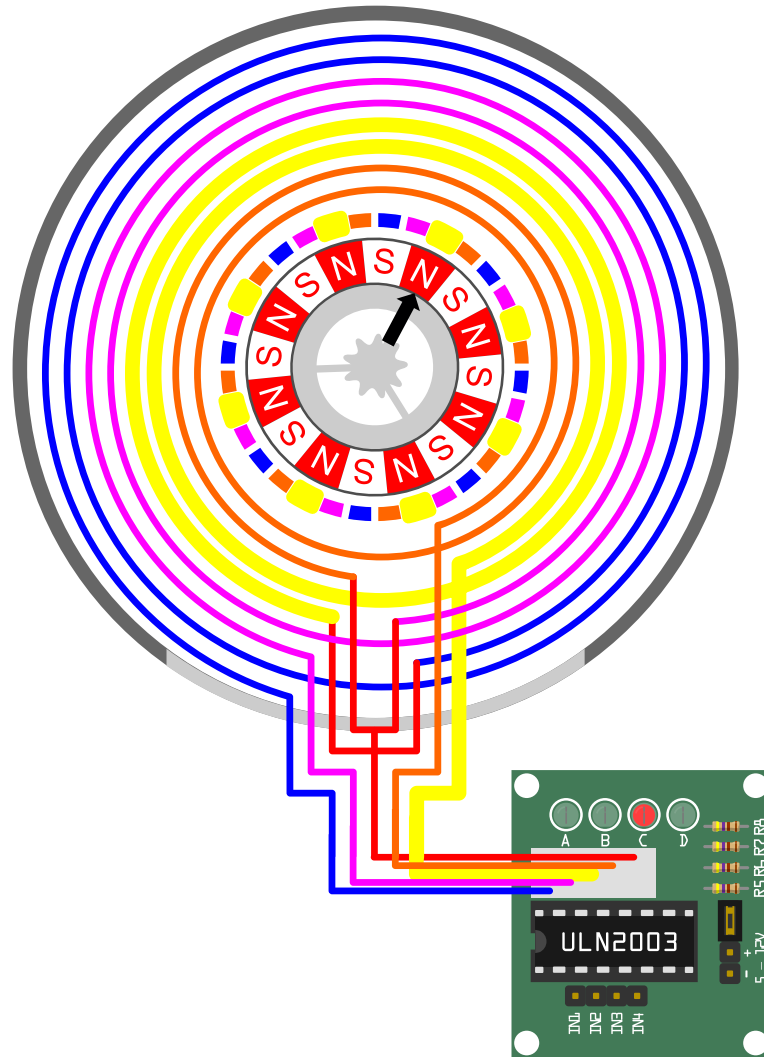
step 1



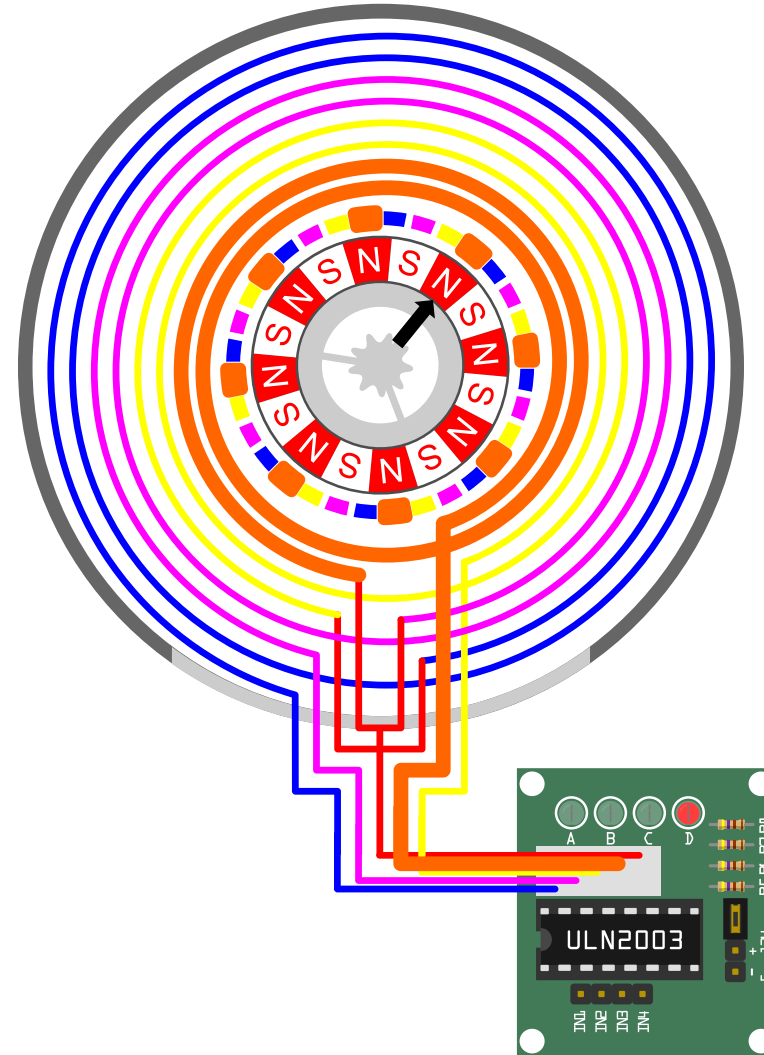
粉色高电平

step 2

Driving Mode – Wave Driving

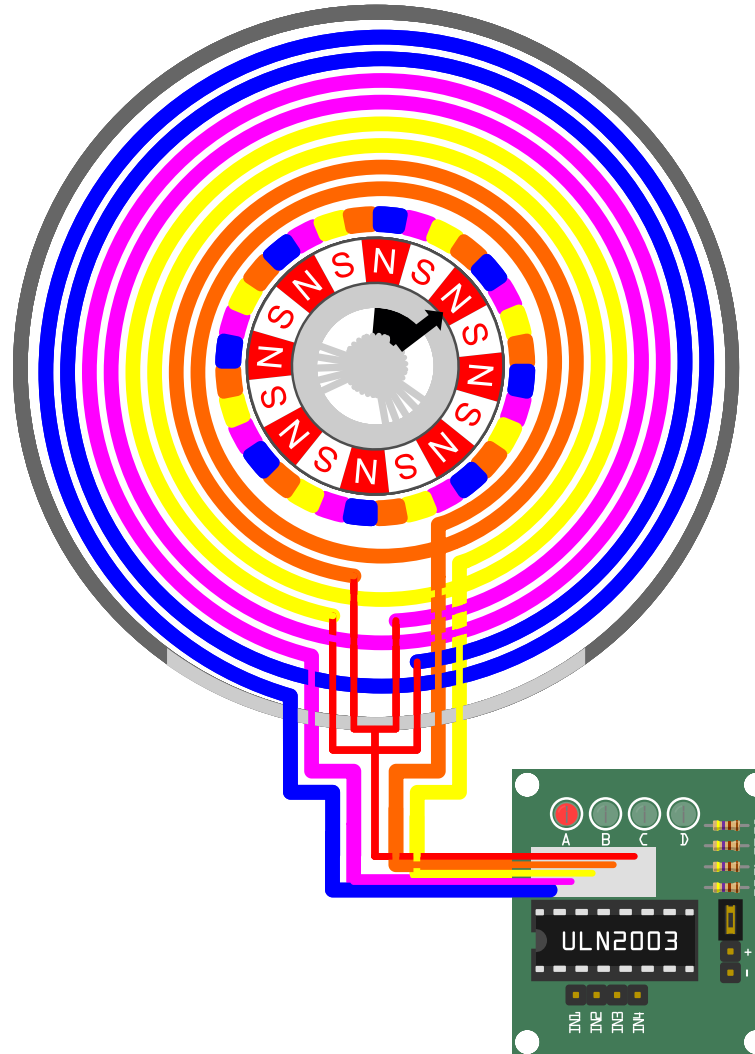


step 3

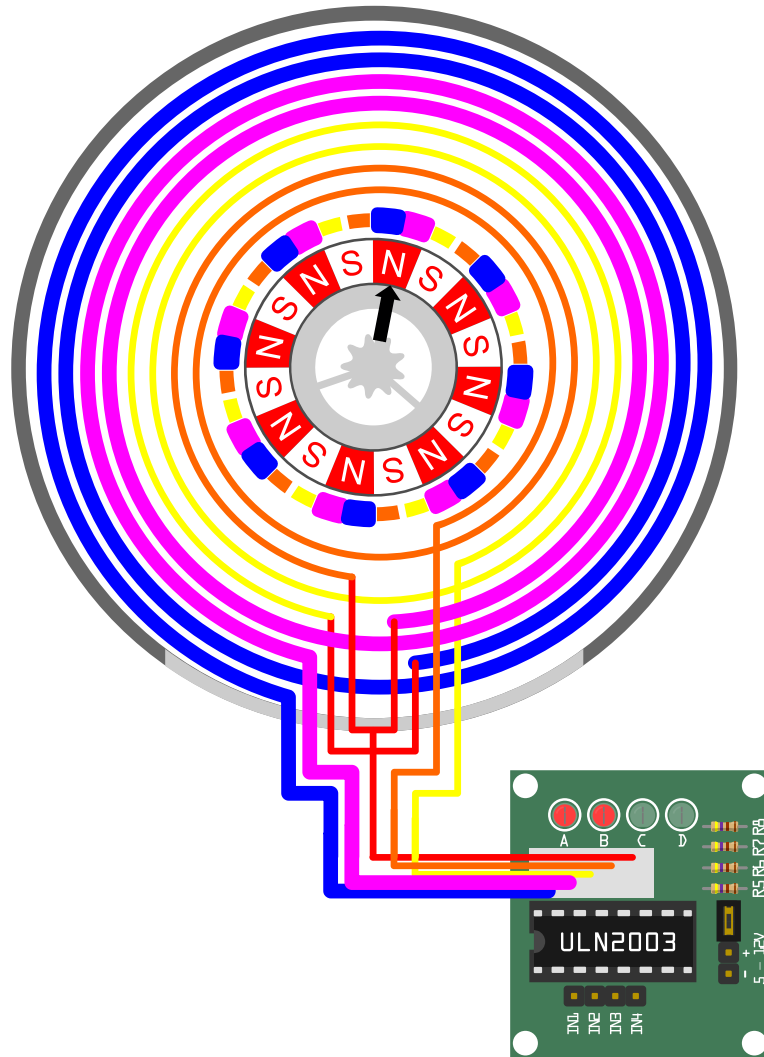


step 4

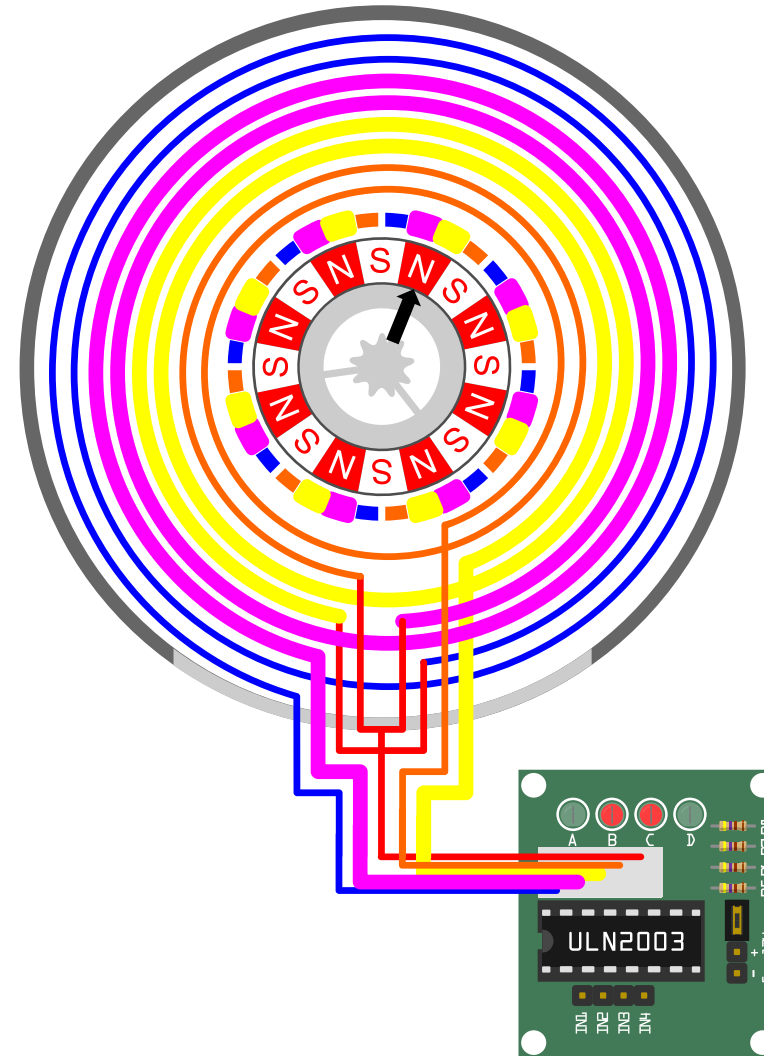
Driving Mode – Wave Driving



Driving Mode – Full Stepping

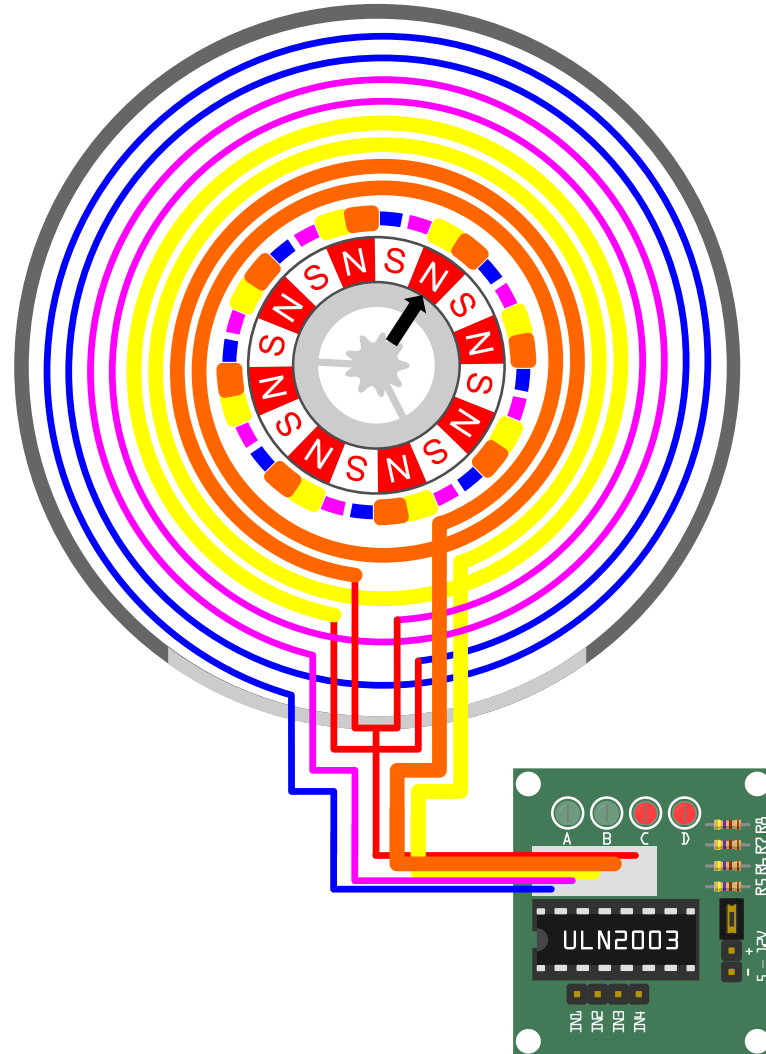


step 1

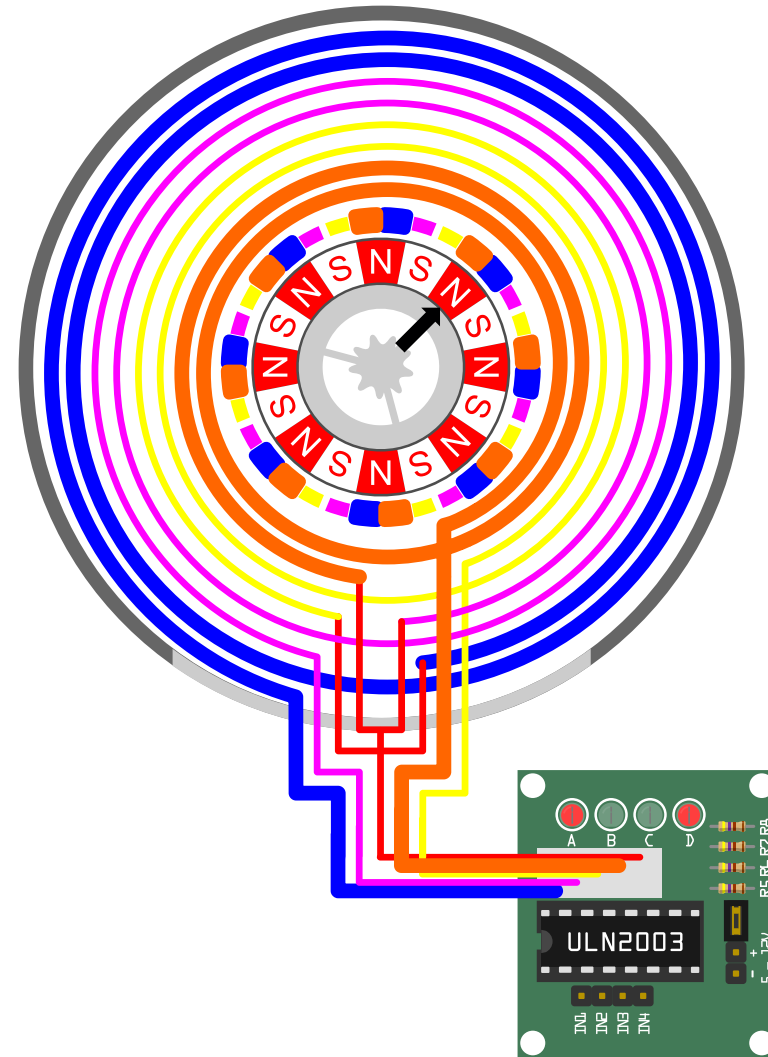


step 2

Driving Mode – Full Stepping

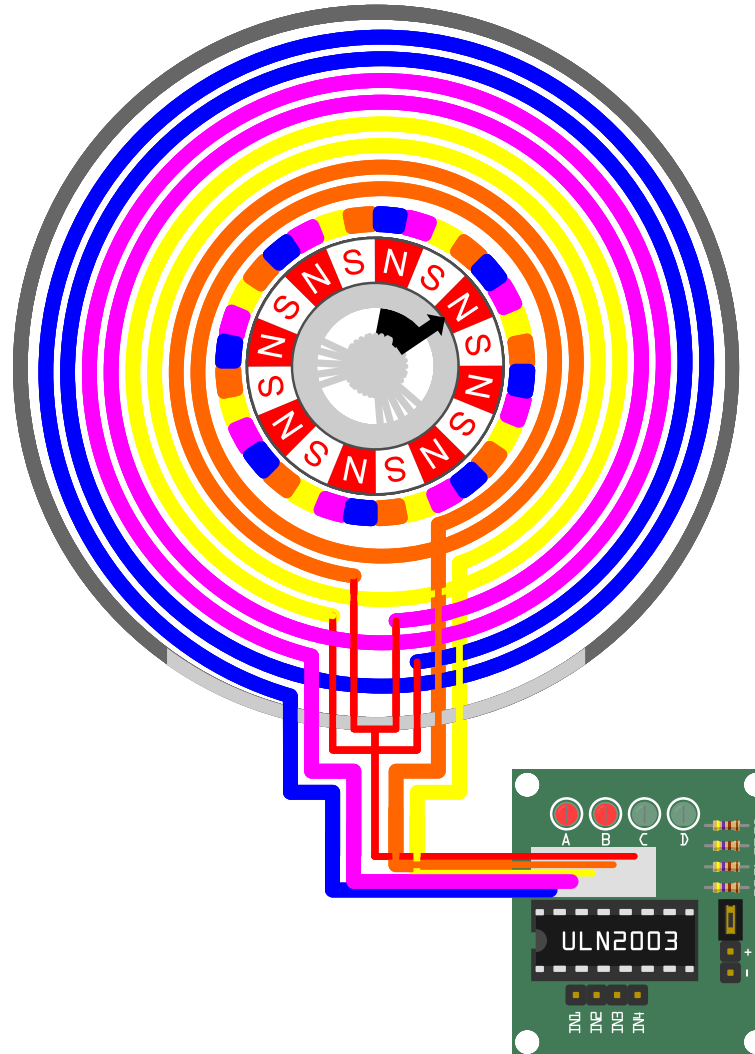


step 3



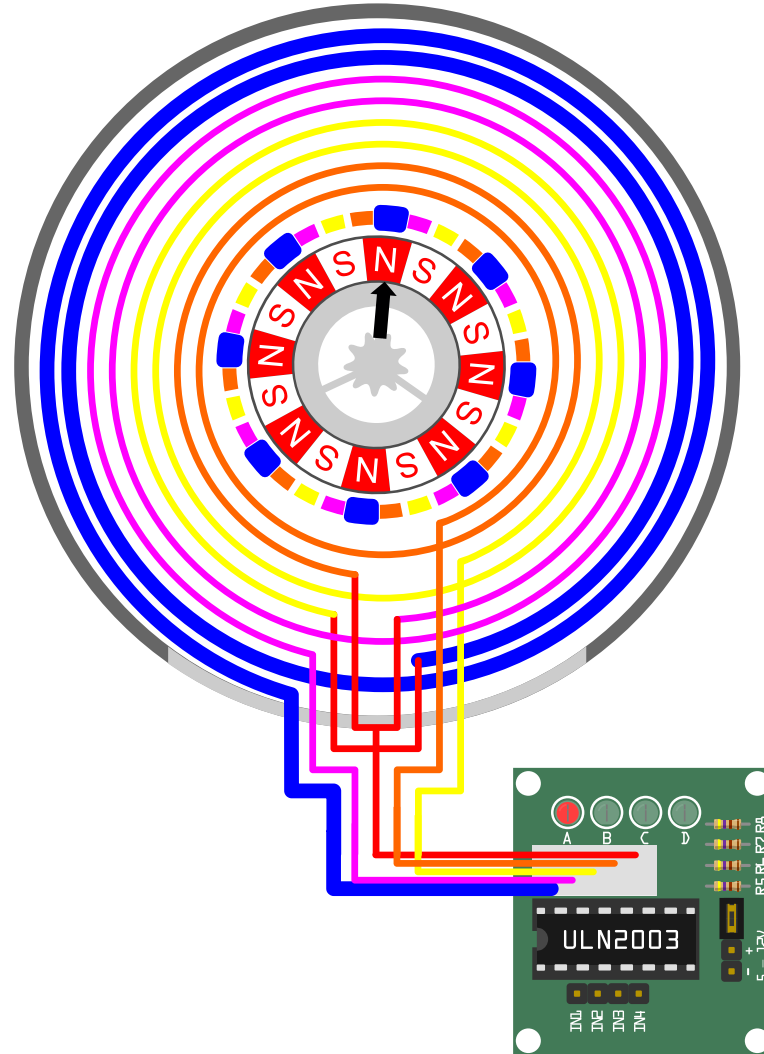
step 4

Driving Mode – Full Stepping

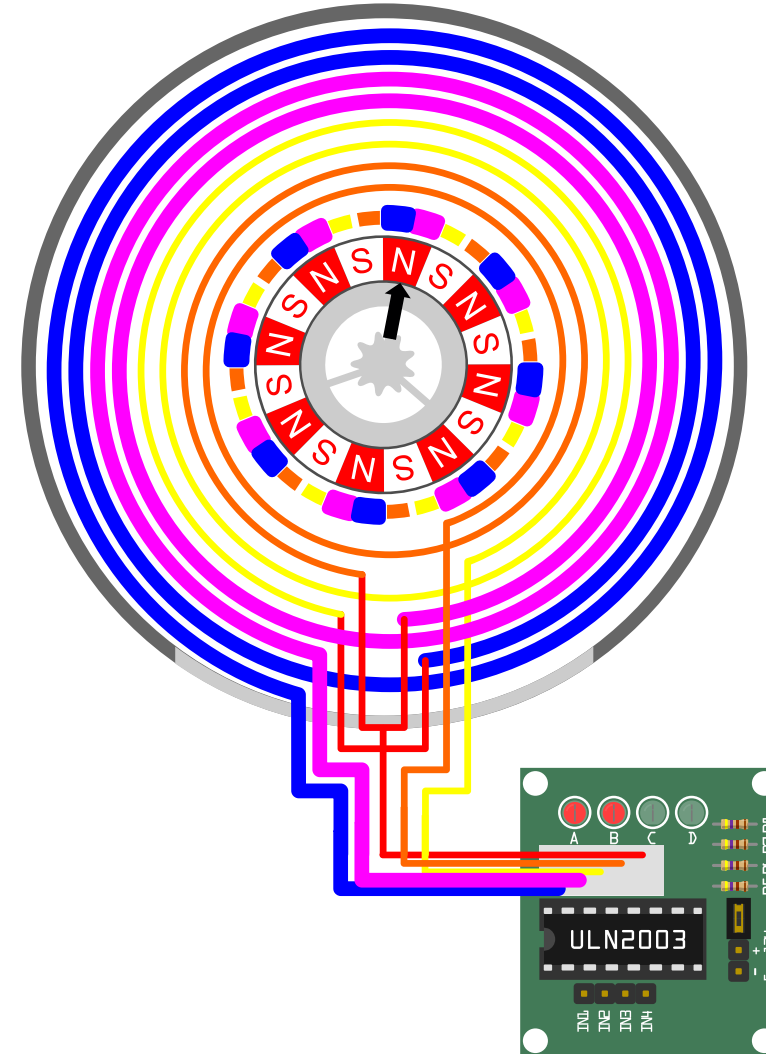


Driving Mode – Half Stepping

更精确

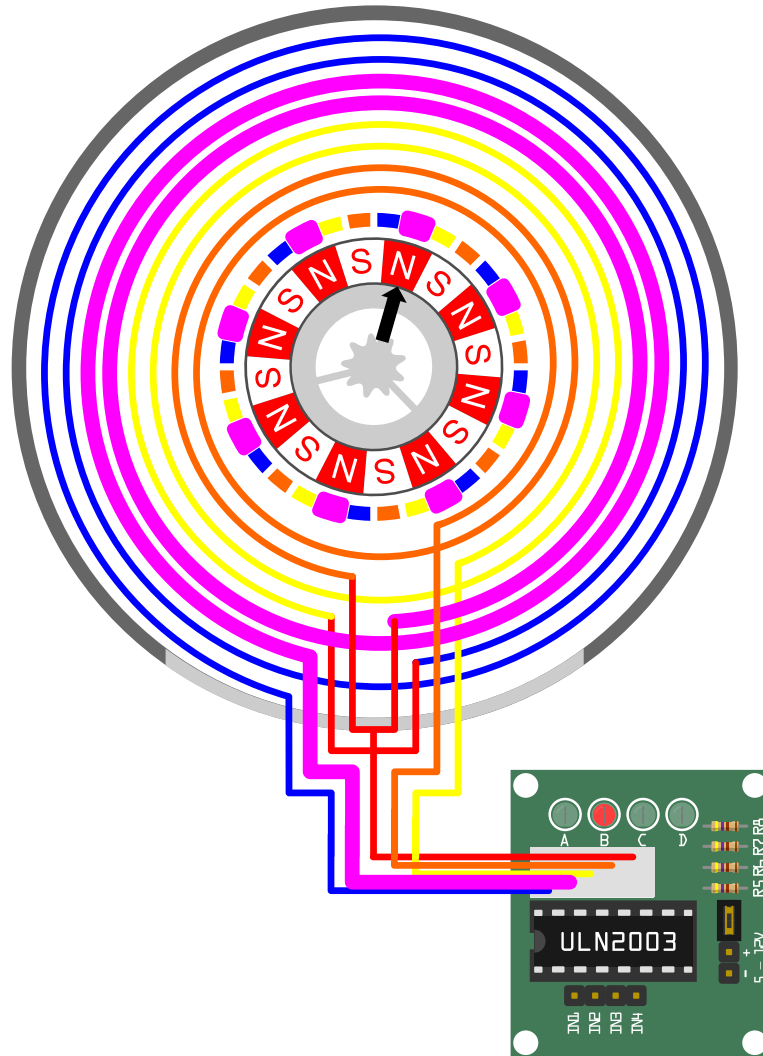


step 1

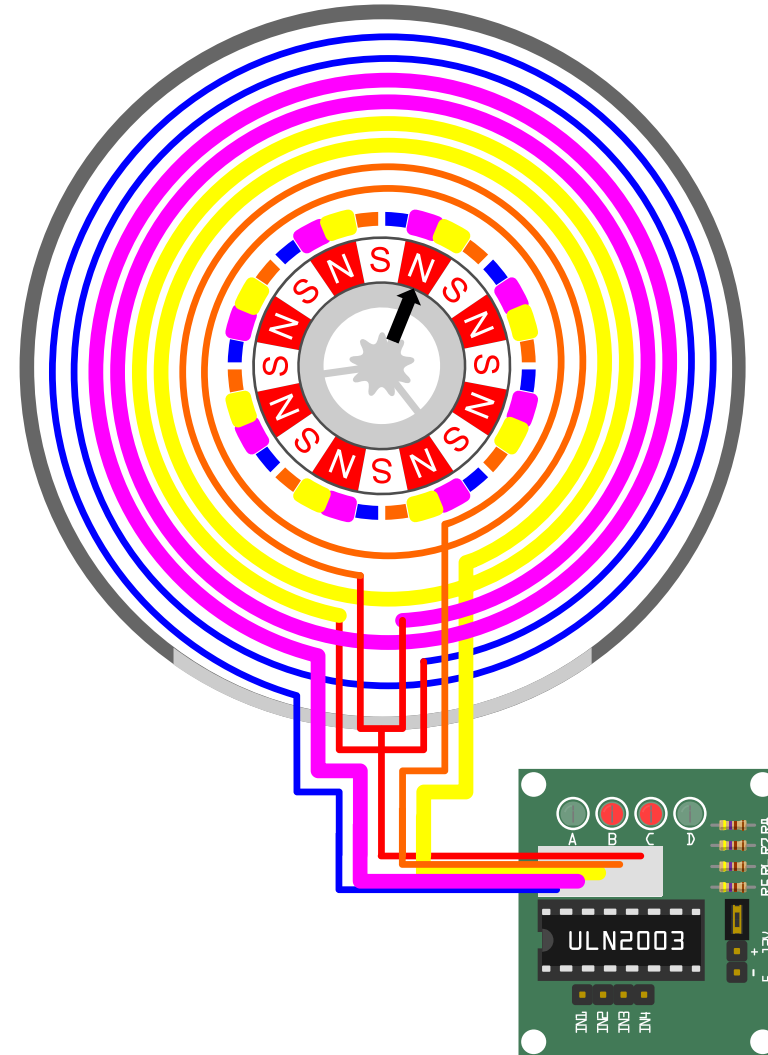


step 2

Driving Mode – Half Stepping

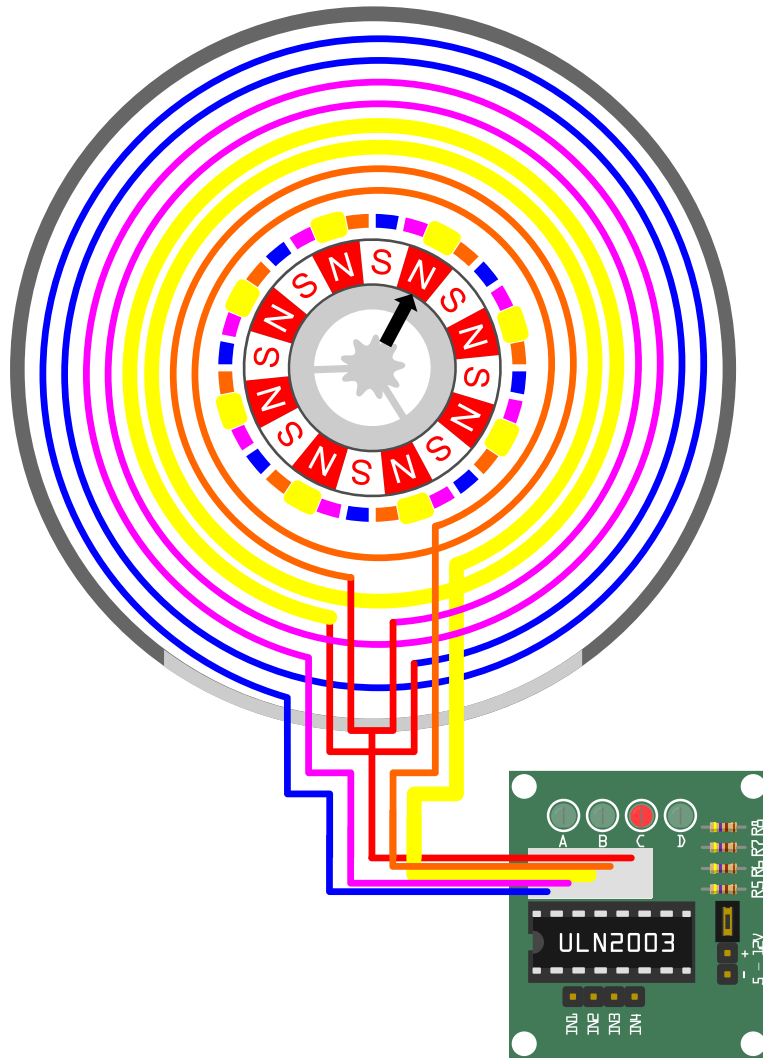


step 3

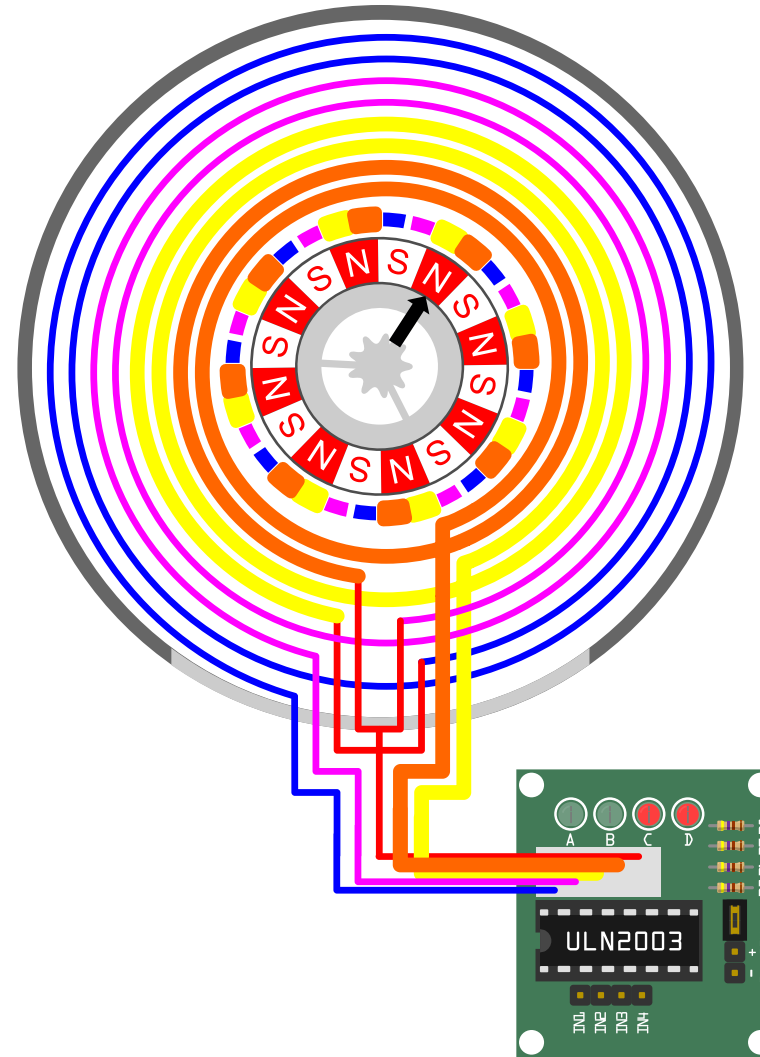


step 4

Driving Mode – Half Stepping

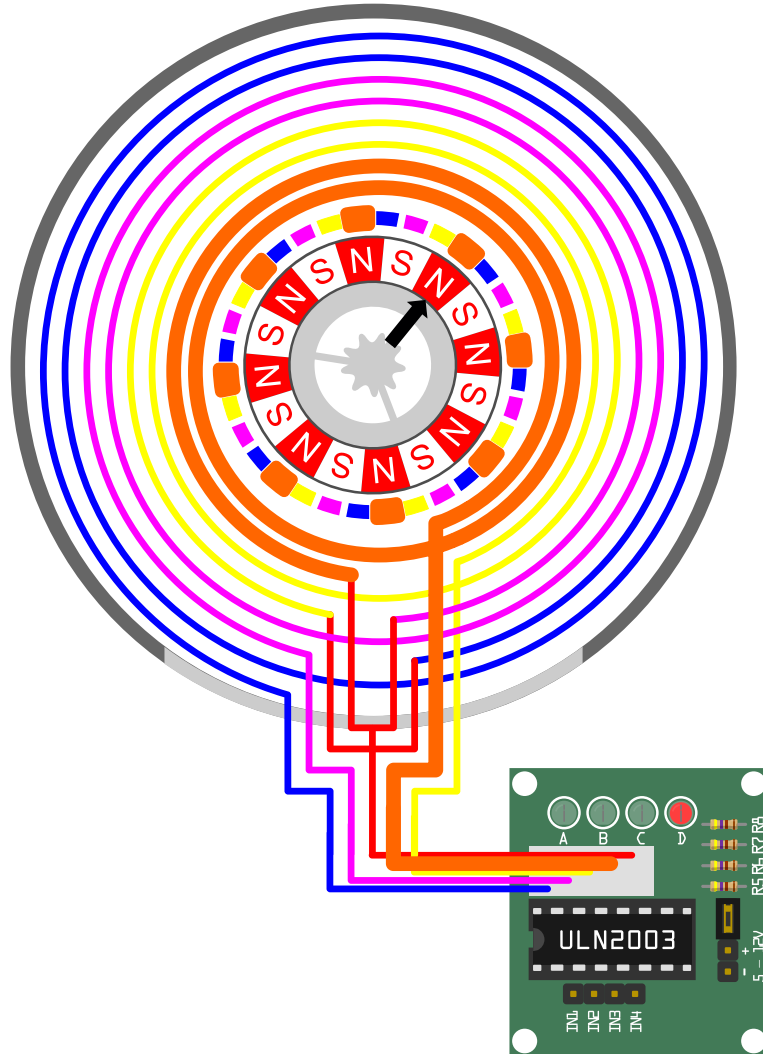


step 5

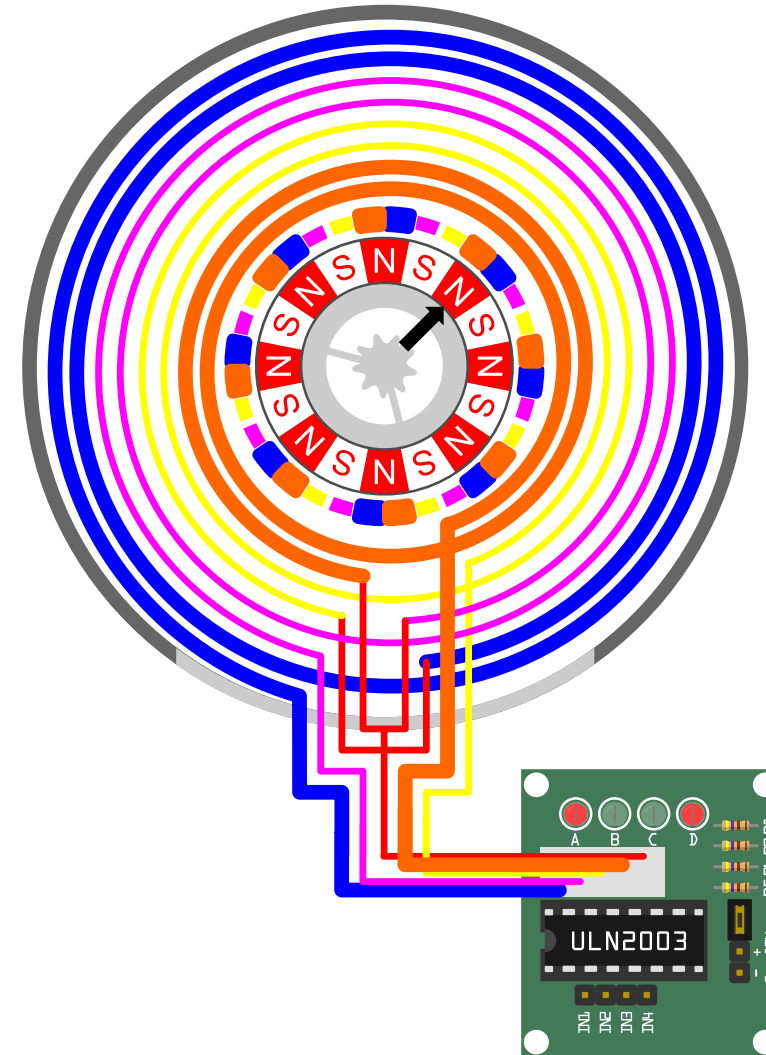


step 6

Driving Mode – Half Stepping



step 7

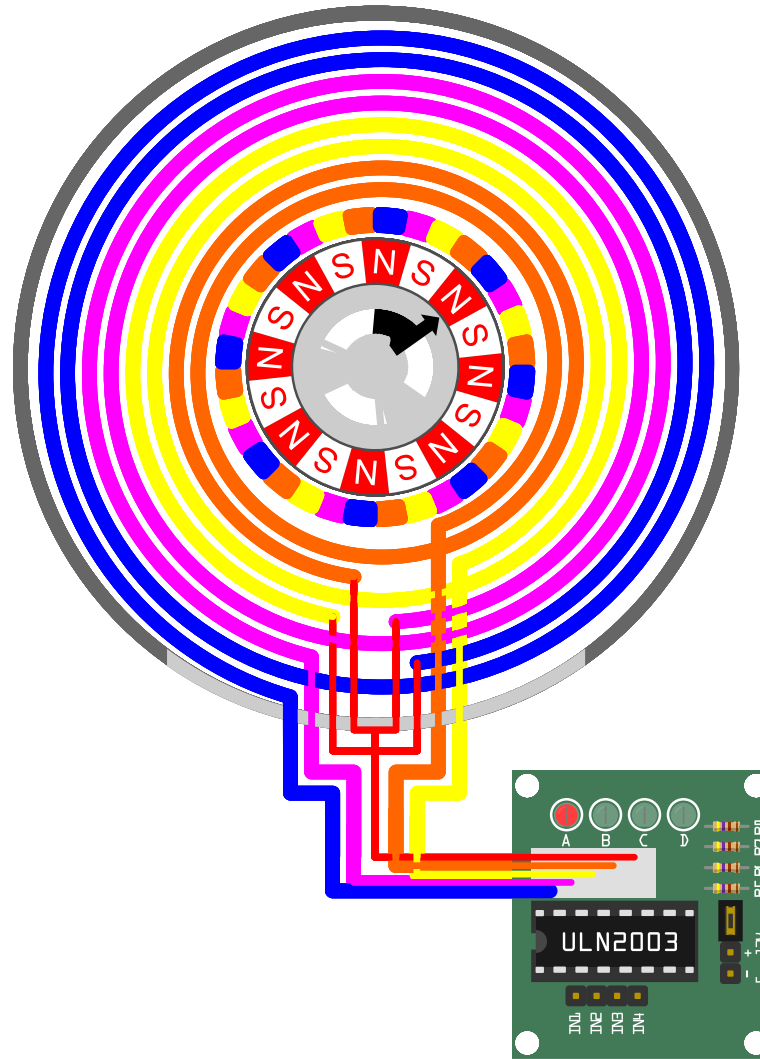


step 8

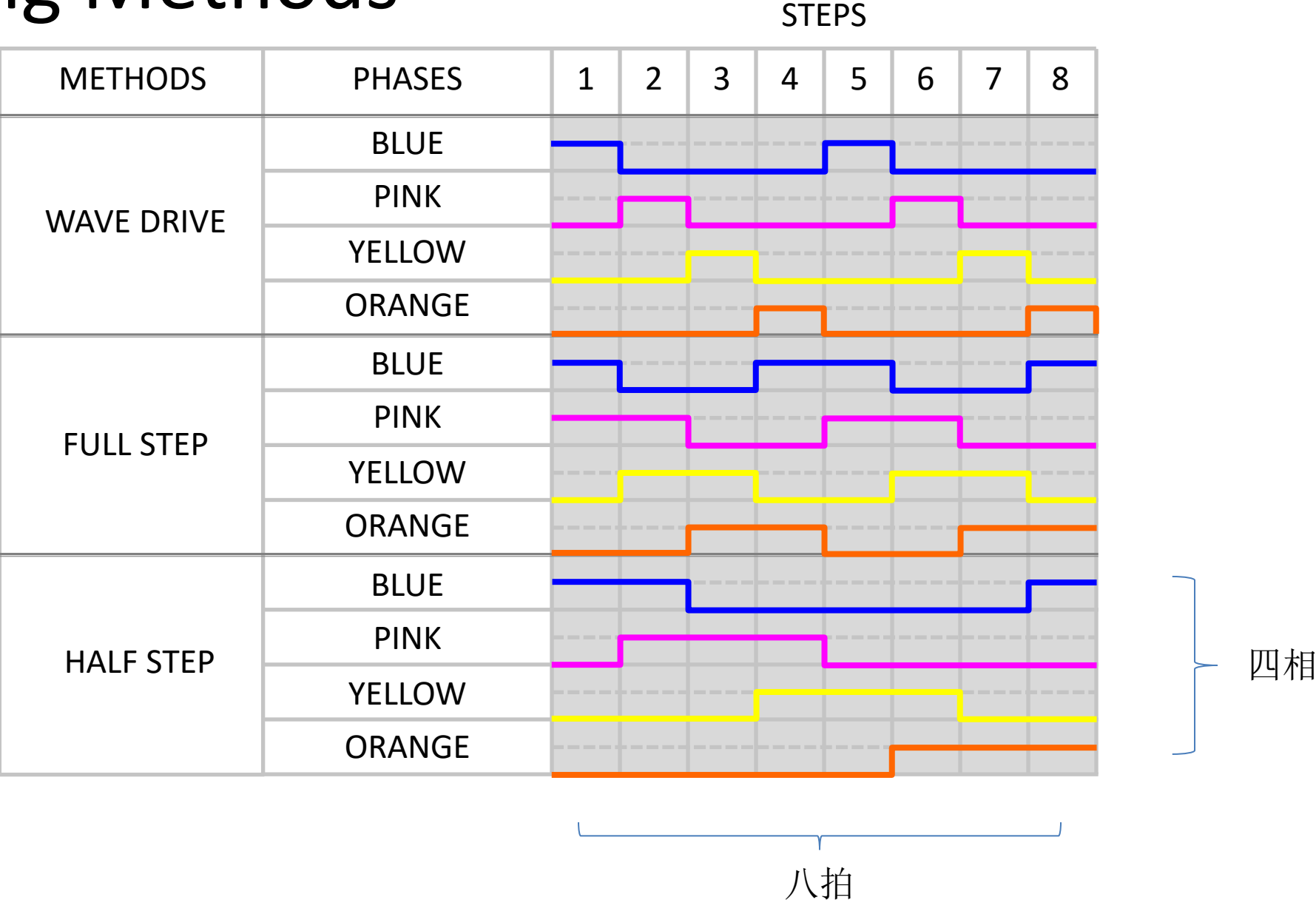
步进 5.625°

如何反转？

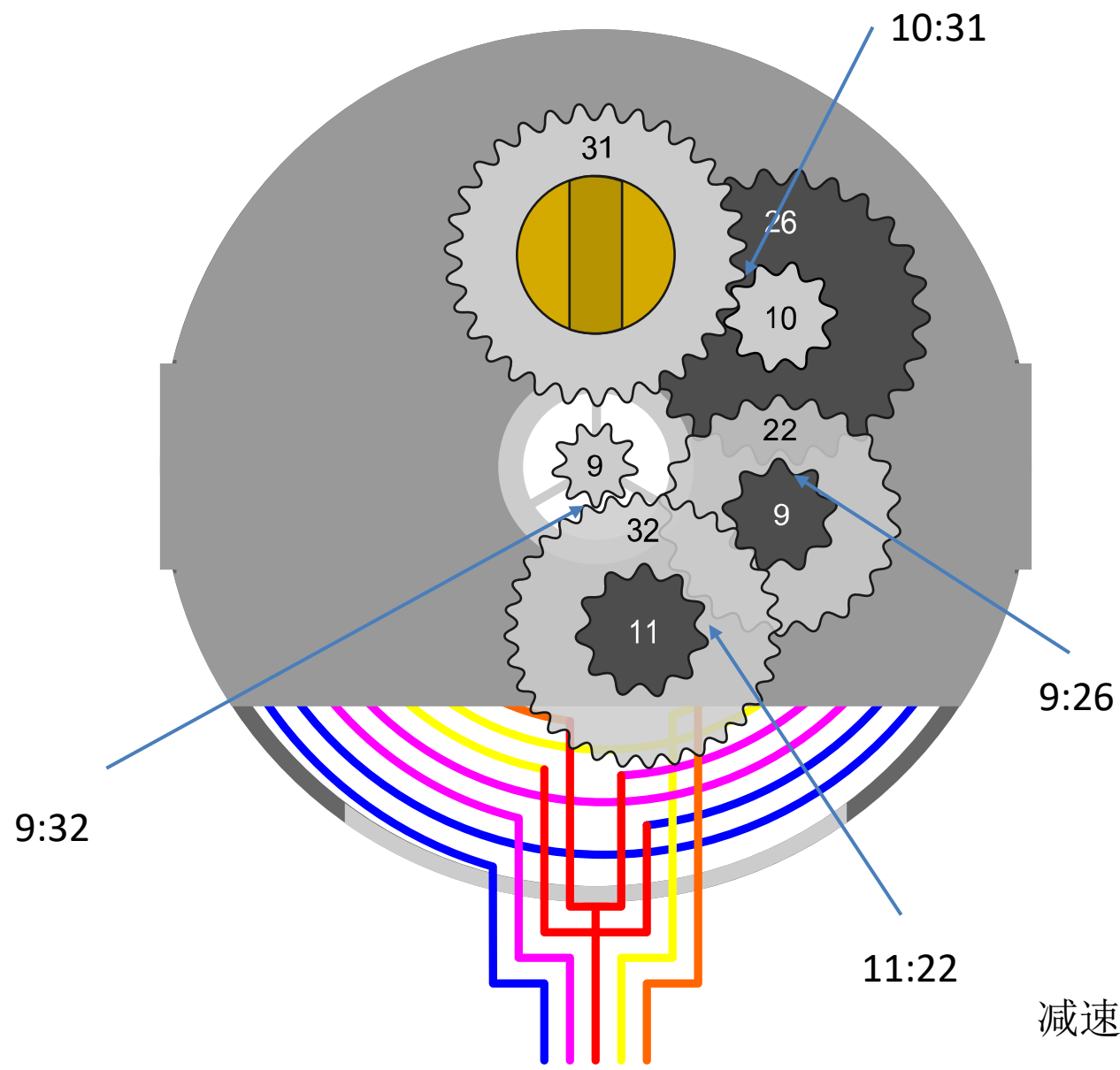
Driving Mode – Half Stepping



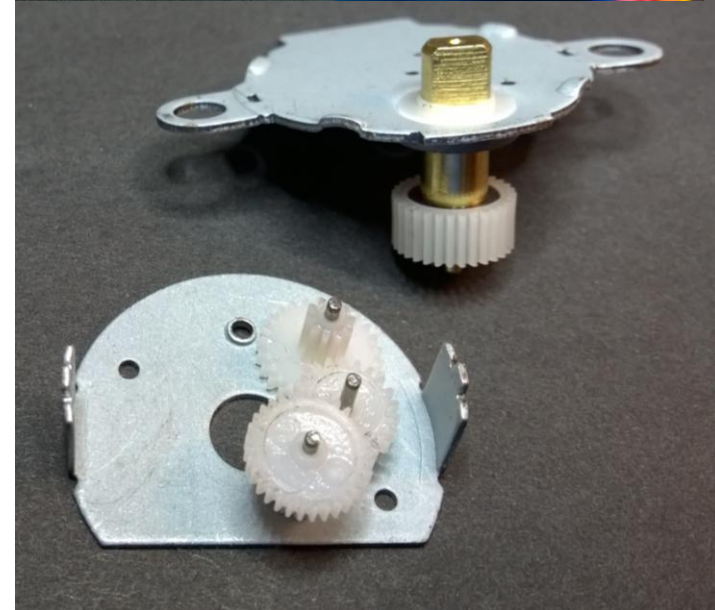
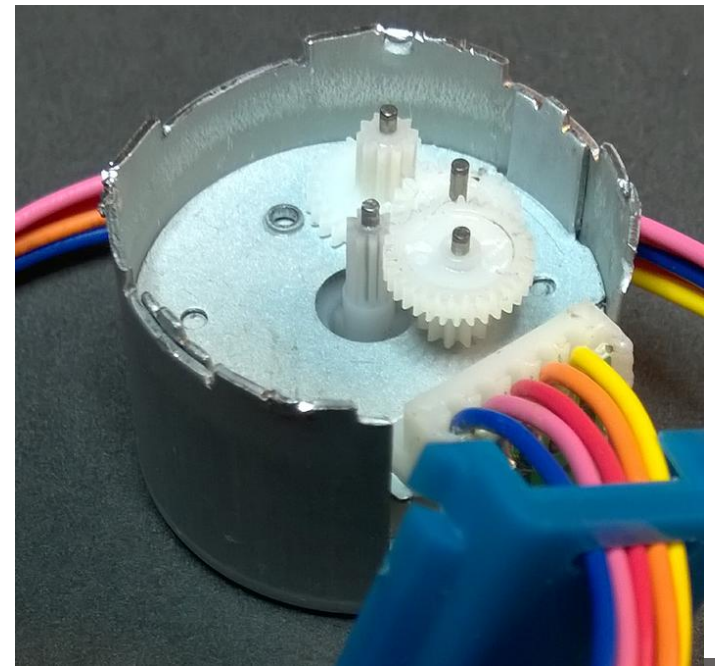
Stepping Methods



Gears

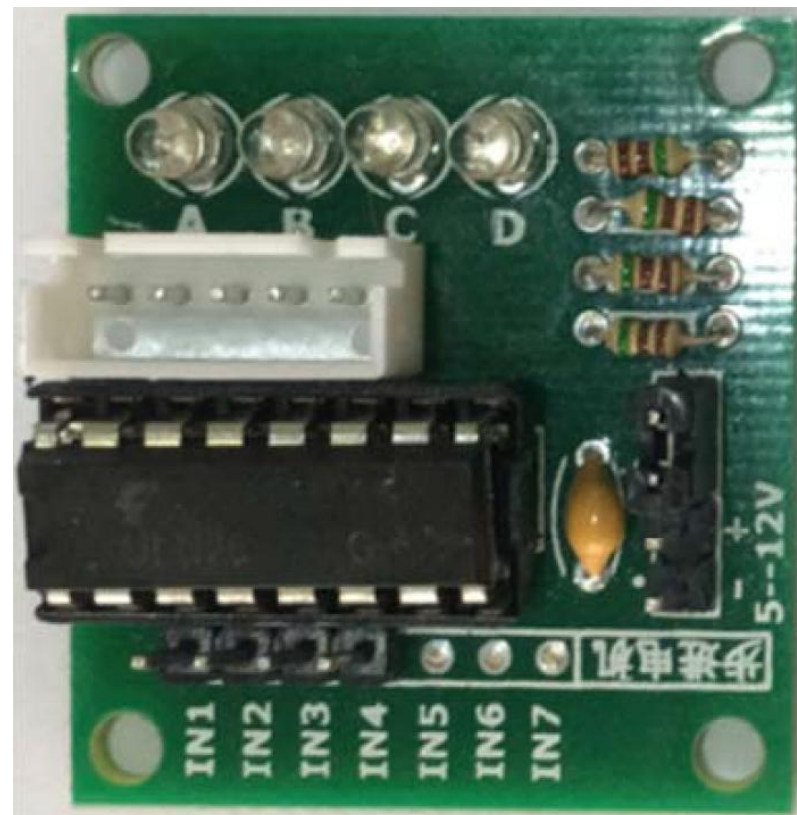
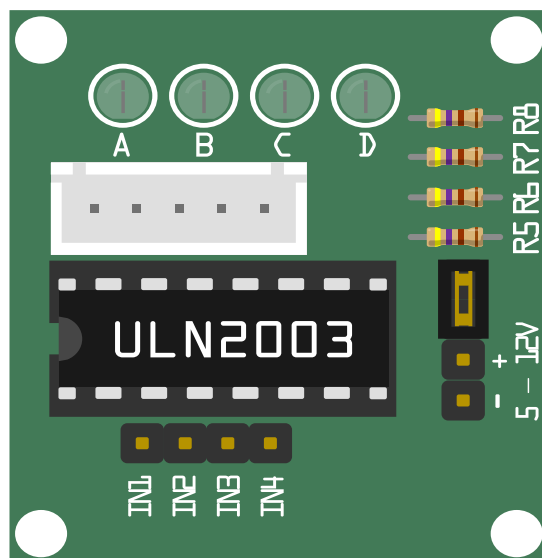


減速比约为 1: 64



Drive Board

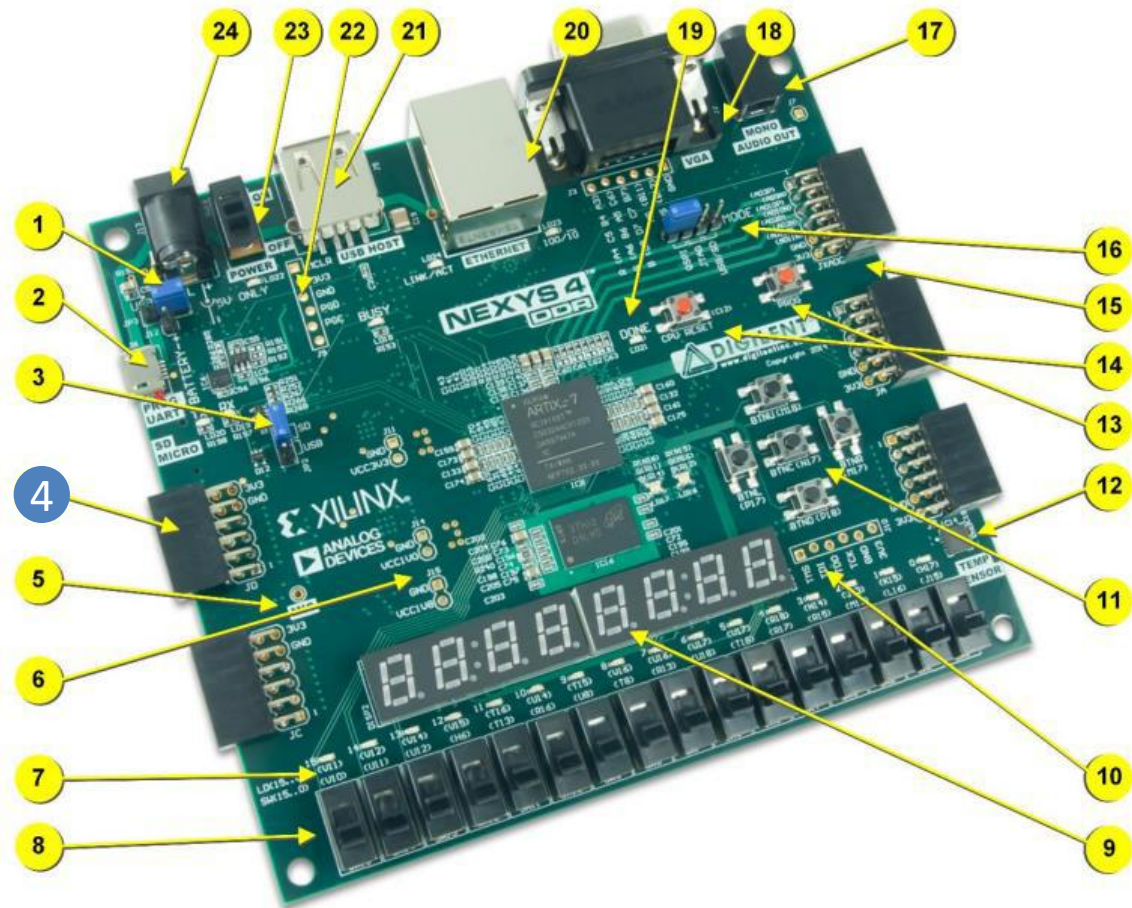
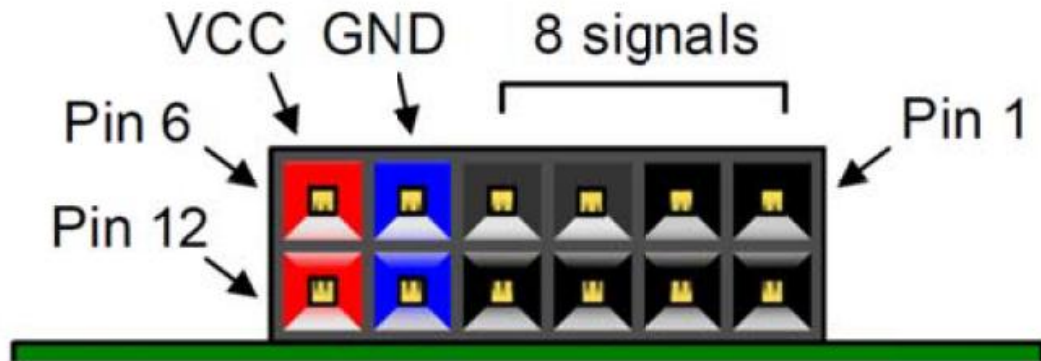
Nexys4 DDR开发板所使用的输入/输出（I/O）标准电压为3.3V，而步进电机需要大于5V的供电电压。因此，需要使用一块驱动板(ULN2003)将3.3V电压转换为5V电压。



Pmod Connector

Nexys4 DDR开发板除了提供了控制板载器件的接口之外，也为我们提供了4个逻辑Pmod (peripheral module)接口和1个模拟Pmod接口以扩展开发板的功能。

Pmod 接口呈 2×6 排列，如图所示，每个Pmod接口提供两个 3.3V 的 VCC 信号（第 6 针和第 12 针）、两个接地信号（第 5 针和第 11 针）以及八个逻辑信号。VCC 引脚和接地引脚最高可承载 1 A 的电流。

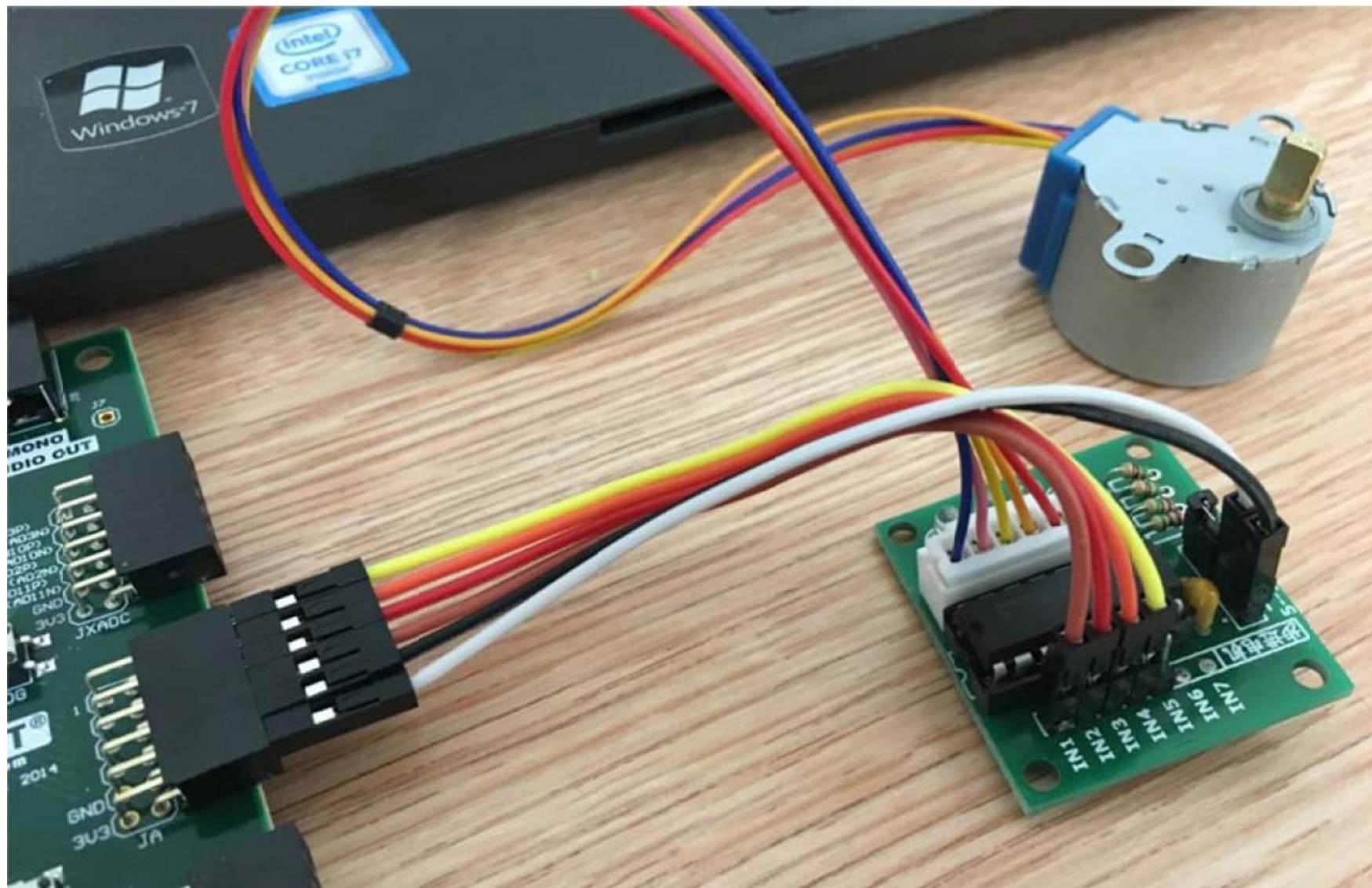


FPGA Pins for Pmod Connector

Pmod JA	Pmod JB	Pmod JC	Pmod JD	Pmod XDAC
JA1: C17	JB1: D14	JC1: K1	JD1: H4	JXADC1: A13 (AD3P)
JA2: D18	JB2: F16	JC2: F6	JD2: H1	JXADC2: A15 (AD10P)
JA3: E18	JB3: G16	JC3: J2	JD3: G1	JXADC3: B16 (AD2P)
JA4: G17	JB4: H14	JC4: G6	JD4: G3	JXADC4: B18 (AD11P)
JA7: D17	JB7: E16	JC7: E7	JD7: H2	JXADC7: A14 (AD3N)
JA8: E17	JB8: F13	JC8: J3	JD8: G4	JXADC8: A16 (AD10N)
JA9: F18	JB9: G13	JC9: J4	JD9: G2	JXADC9: B17 (AD2N)
JA10: G18	JB10: H16	JC10: E6	JD10: F3	JXADC10: A18 (AD11N)

FPGA - Driver - Motor

FPGA开发板、驱动板和电机的示例连接如图所示。请注意，驱动板上的“+”和“-”引脚需连接至FPGA开发板所选Pmod连接器的3V3（3.3V电源）和GND（接地）引脚。



Assignment 1

请仔细阅读数据手册中关于Pmod Connectors的部分。

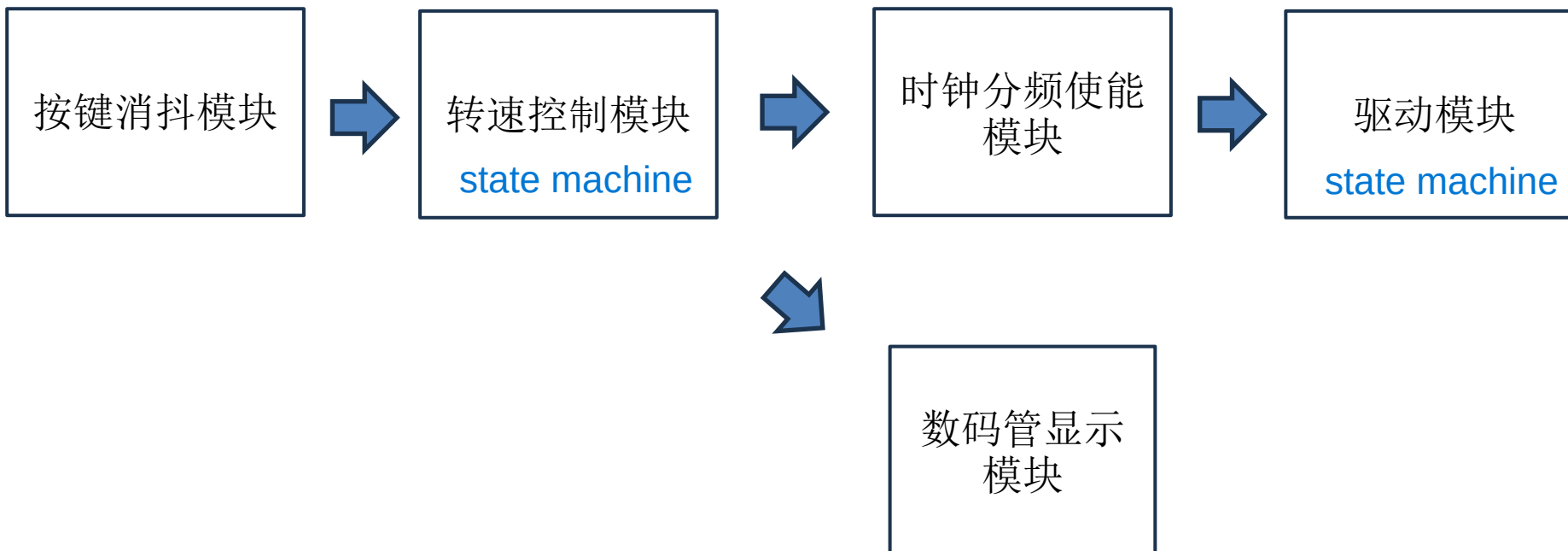
请设计步进电机的驱动电路并实现以下功能：

1. 请使用滑动开关SW0实现步进电机启动和停止功能；
2. 请使用滑动开关SW15实现步进电机正转和反转；
3. 请使用BTNU和BTND实现增速和减速（0-9转/分钟）；
4. 请使用数码管显示转速（1位即可）；
5. 推荐的模块设计见下一页。

本题需要验收，请演示以上所有功能。

Assignment 1

- 结构框图



Assignment 2

- 乘法器

请针对理论课件中设计的两种基于8位加法器的乘法器设计进行仿真 (combinational design & repetitive-addition design)。

请从器件利用率、延迟等方面比较不同设计的性能。

Combinational design: Lecture Note Part II-1, P185~P191

Repetitive-addition design: Lecture Note Part II-3, P241~259

Assignment 3

- 乘法器

参考理论课件中pipelined design 8位乘法器，设计一个n位的乘法器流水线设计，其中n是1到9之间的正整数。对该设计进行仿真，指明延迟和吞吐量，并将其与Assignment 2中的设计进行比较。

Pipelined design: Lecture Note Part II-6, P322~330

Assignment 2&3

1. 请编写testbench文件对设计进行仿真；
2. 请展示post implementation simulation results；
3. 请展示schematics以及其他综合后的各项数据；
4. 请对结果进行分析。

Assignment2和3合并提交报告，包括：题目、关键代码（合理注释）、结果、分析等。